Lavya Jain

+91-8218476656 | lavyajain.dev@gmail.com | Portfolio | LinkedIn | GitHub

Education

Dr. D.Y. Patil Institute of Technology, Pune — Aug 2023 – July 2027
B.E – Information Technology CGPA: 8.14/10 — *Pune, Maharashtra*

St. Joseph's Academy, Dehradun — 2022
Class XII (ISC) – 90.2% — *Dehradun, Uttarakhand*

St. Joseph's Academy, Dehradun — 2020
Class X (ICSE) – 91.2% — *Dehradun, Uttarakhand*

Projects

Sanrakshan CTC | *C++17, Crow, WebSockets, JavaScript* | GitHub | Demo — Jan 2026 – Mar 2026

- Engineered a multithreaded railway traffic control simulator where every train executes independently using **std::thread**, coordinated through mutex-protected shared track resources.
- Implemented collision avoidance using **non-blocking try_lock()**, eliminating hold-and-wait deadlocks while dynamically rerouting trains through graph-based path planning.
- Built a live WebSocket telemetry pipeline with a browser-based SCADA dashboard providing real-time train tracking, signal visualization, emergency stop controls, and infrastructure monitoring.
- Designed a concurrent deadlock demonstration mode that intentionally recreates Dining Philosophers-style resource contention for educational visualization of operating system synchronization concepts.

VAJRA | *C++, React, Three.js, Node.js, ZeroMQ, WebSockets* | GitHub | Demo — Mar 2026 – Apr 2026

- Architected a cyber-physical power grid simulator that models electrical load propagation, cascading failures, and infrastructure attacks through a deterministic C++ simulation engine.
- Developed a ZeroMQ-to-WebSocket bridge enabling sub-second bidirectional communication between a compiled C++ backend and an interactive React command center.
- Built a 3D digital twin using React Three Fiber with real-time telemetry overlays, threat visualization, AI voice alerts, and operator-focused SCADA interfaces.
- Implemented simulated industrial cyberattacks including False Data Injection, Packet Spoofing, and Volumetric DDoS with automated anomaly detection and node isolation strategies inspired by MITRE ATT&CK for ICS.

VeriFind | *React Native, Node.js, MongoDB, Solidity, IPFS* | GitHub | Demo — Aug 2025 – Feb 2026

- Designed a blockchain-backed ownership verification platform that establishes immutable ownership history for physical electronic devices using ERC-721 smart contracts.
- Implemented a gasless meta-transaction relayer allowing users to interact with blockchain assets without holding cryptocurrency while preserving cryptographic ownership verification.
- Built a React Native mobile application integrated with MongoDB and IPFS for real-time device registration, ownership transfer, and decentralized metadata storage.
- Developed secure ownership validation that restricts sensitive actions such as transfer, lost-device reporting, and recovery exclusively to the wallet cryptographically linked to the registered device.

Technical Skills

Languages: C++, JavaScript, TypeScript, Java, SQL
Frameworks & Libraries: React, React Native, Node.js, Express.js, Three.js, React Three Fiber, Solidity, Ethers.js, Crow, Tailwind CSS
Databases & Storage: MongoDB, PostgreSQL, IPFS
Developer Tools: Git, GitHub, Linux, Docker, ZeroMQ, WebSockets, Postman, Vercel
Core Concepts: Data Structures & Algorithms, Operating Systems, Computer Networks, Multithreading, Concurrent Programming, Distributed Systems, Blockchain

Achievements

- Winner, **Hash-It-Out Hackathon 2026**, Dr. D.Y. Patil Institute of Technology.
- Finalist, **Hackathon Vortexa 2025**, presenting an AI-powered financial strategy platform.
- Built and demonstrated multiple real-time systems spanning concurrent simulation, blockchain ownership verification, industrial control visualization, and cyber-physical infrastructure defense.